
Intracranial Regulation & Head Trauma

Learning Objectives

1. Define characteristics of clients head injuries and traumatic brain injuries.
 2. Discuss principles of intracranial regulation as they relate to cerebral blood flow.
 3. Compare types of head injury by mechanism of injury and clinical manifestations.
 4. Describe the interprofessional and nursing management of the client with increased intracranial pressure.
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Monro-Kellie Doctrine

- **Monro-Kellie doctrine:** 3 components must stay at a relatively constant volume within the closed skull to maintain function
 - Three ways to maintain a normal ICP:
 - Change CSF volume
 - Alter blood flow via vasoconstriction or vasodilation
 - Dura stretches or brain tissue compresses
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Cerebral Pressures

Intracranial pressure (ICP): hydrostatic force measured in the CSF compartment

- *Normal ICP: 5-15mmHg*

Cerebral perfusion pressure (CPP): pressure needed to maintain adequate blood flow in the brain

- *CPP = MAP - ICP*
- *Normal is 60-100mmHg; ideal is 60-70mmHg*
- Less than 50mm Hg: tissue ischemia & death; less than 30mm Hg: incompatible with life

Will need to calculate CPP, and know if it's normal.
Co2 increases blood flow to the brain
Fever can increase cerebral blood flow.
Neuroprotective to keep patients a little cooler.
Coughing increases ICP

Increased ICP

- Result of an increase in any or all of the 3 components (brain, blood, CSF) within the skull
- Matters because CPP decreases - impairs autoregulation
- Sustained increased ICP results in brainstem compression and brain herniation - irreversible

Once we herniate, there's nothing else we can do for this patient.

Head Injuries

- Broad term that refers to any trauma to the scalp, skull, face or brain
- Most common causes: falls, motor vehicle accidents
- Other causes: firearms, assaults, sports-related trauma, war-related injuries
- High potential for poor outcomes
- Severity:
 - Mild (GCS 13–15) – Often a concussion
 - Moderate (GCS 9–12)
 - Severe (GCS 3–8)

Most likely to die at time of injury, two hours after, or 3 weeks after

Head Injuries

Focal injury:

- Contusions
- Coup-contrecoup injury
- Lacerations
- Diffuse injuries:
 - Concussions
 - Diffuse axonal injuries

Diffuse are tougher to diagnose with imaging
Diffuse axonal cant be seen on ct or mri

Skull Fractures

Types: linear, depressed,
open, or basilar

Basilar have specific manifestations

Skull Fractures

- Signs of a skull fracture:
 - Battle's sign
 - Raccoon eyes
 - Clear rhinorrhea/otorrhea - Halo's sign
 - Hemotympanum

Battle sign – post auricular bruising.

Traumatic Brain Injuries

- More severe head injury involving damage to brain function caused by an external force
 - Includes evidence of brain involvement
 - Loss/alteration of consciousness
 - Amnesia
 - Neurological deficits
 - Cognitive, emotional, or behavioral changes
 - Seen on imaging
 - Can be caused by mechanical, thermal, chemical, electrical, radiation
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Emergency Care

Priorities

- Stabilize C-spine
- Airway and oxygenation
 - GCS less than 8 → intubate
- IV access
- Control bleeding
- Rapid neuro assessment (GCS)
- Prepare for CT scan

Diagnostics

- CT scan: First-line for rapid assessment of bleeding or swelling
- MRI: Detailed soft tissue assessment
- Cervical spine x-ray: Rule out neck injuries

Goals of care: stabilize; prevent secondary injury

Assessing ICP

- Biggest problem in head injuries is abnormal ICP
- Manifestations of increased ICP:
 - Changes in LOC
 - Vital signs
 - Pupils
 - Positioning
 - Other neurological symptoms
- Diagnostic testing: MRI, EEG, CT angiogram, LICOX catheter

Both pupils fixed and dilated is ominous.
Eye drops, previous eye surgeries?
Problematic positioning – decerebrate, dicordiicate.

Interprofessional Management

- Breathing
 - Mechanical ventilation
 - $\text{PaO}_2 \geq 100 \text{ mmHg}$
- Maintain fluid balance - goal is euvoemia
- Maintain systolic arterial pressure between 100 and 160 mm Hg
- ICP monitoring
 - Maintain CPP > 60 mmHg
- Nutrition - PO or enteral

Breathing is big priority, intubating, and suctioning increases icp
Can tolerate icp up to 20

Medications

Goal is to promote cerebral perfusion

- Diuretics:
 - Mannitol
 - Hypertonic saline (3% normal saline)
- Corticosteroids (**contraindicated in head injuries**)
- Barbiturates (i.e. phenobarbital)
- Antiseizure medications (i.e. phenytoin)
- Sedatives - propofol is preferred; avoid benzodiazepines

Mannitol – osmotic diuretic pulls fluid from cerebral tissue and drains it from the brain.

No steroids in head injuries

Barbiturates to decrease cerebral metabolism

Kepra also commonly used.

Propofol is preferred – has a short half life. Easy to neuro assessments quickly.

Benzo's bad, long half life – also greater risk of hypotension.

Nursing Care

- Assessment
 - Respiratory assessment
 - Caution with suctioning
 - Glasgow Coma Scale
 - Neurological assessment
 - Is & Os
- Monitoring:
 - Maintain systolic arterial pressure between 100 and 160 mmHg
 - Maintain ICP between 5-15 mmHg
 - ABGs

15 is normal, 8 we are in big trouble

ABGs- co2 monitoring, make sure o2 concentration is good.

Nursing Care

- Interventions
 - Positioning
 - Place NG tube to low intermittent suction - **contraindicated with facial trauma & skull fractures, place OG instead**
 - Monitor urine output - why?
 - Restraints may be needed to allow care to continue
 - Decrease environmental stimuli
 - Minimize any interventions or procedures that may increase ICP

Little bit upright, head in neutral position – allows best venous return
Slow position changes are key
SIADH or DI in a head injury patient.

Herniation

- “Brain death”
- Criteria include coma, lack of brainstem reflexes, & apnea
- Diagnosis made by a physician via neurological exam
 - Eye assessment - oculoccephalic reflex (“doll’s eye reflex”), oculovestibular reflex

Done twice, 12 hours apart to assess for brain death.

Organ Donation

- **Do not initiate discussions with the family**
 - Notify organ procurement if:
 - Family/hospital considering withdrawal of care
 - Comfort cares discussed
 - Change to DNR status
 - Brain death is confirmed (DBD)
 - Loss of one or more brainstem reflexes
 - Comfort cares discussed
 - Care is withdrawn and death follows (DCD)
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